MZ2POL-07: Validation and Troubleshooting

> **Series:** MZ2POL — Management Zone Migration: Segments, Policies, Workflows | **Notebook:** 8 of 14 | **Created:** December 2025 | **Last Updated:** 10/13/2026

Overview

This notebook provides comprehensive validation procedures and troubleshooting guidance for your migration from Management Zones to Policies, Boundaries, and Segments. Use this guide to verify your migration success and resolve common issues.

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Prerequisites

- Completed migration (MZ2POL-06)
- Admin access to Dynatrace
- Test user accounts for validation

Learning Objectives

By the end of this notebook, you will:
1. Be able to validate migration completeness
2. Know how to troubleshoot common access issues
3. Understand ongoing maintenance requirements
4. Have queries for monitoring access health

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1. Migration Validation Checklist

Component Validation

Component	Validation	Status
User Groups	All groups created and populated	☐
Policies	Policies bound to all groups	☐
Boundaries	Boundaries applied where needed	☐
Security Context	All entities have security context	☐
Segments	Segments created for all filtering needs	☐
Dashboards	Updated to use segments	☐
Alerts	Updated (when segment support available)	☐

Access Validation Matrix

For each user type, verify:

User Type	Can Access	Cannot Access	Verified
Frontend Dev	Frontend services	Backend services	☐
SRE	All production	Development	☐
Viewer	View dashboards	Edit settings	☐

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2. Security Context Validation

Check Coverage

```
// Validate: All services have security context
fetch dt.entity.service
| summarize
    total = count(),
    withContext = countIf(isNotNull(dt.security_context)),
    missingContext = countIf(isNull(dt.security_context))
| fields total, withContext, missingContext,
         coveragePercent = round(100.0 * withContext / total, decimals: 2)

// Note: dt.entity.* is deprecated, but this is a migration-ASSESSMENT query — it inspects
// the classic constructs this migration replaces, so keep the classic query above:
//   - dt.security_context is an ARRAY on Smartscape nodes (empty [] when unset, not null),
//     so isNull / isNotNull(dt.security_context) does not carry over — a Smartscape rewrite
//     would miscount coverage.
```

```
// Find services missing security context
// These need to be addressed for complete migration
fetch dt.entity.service
| filter isNull(dt.security_context)
| fields entity.name, tags, managementZones
| sort entity.name asc
| limit 50

// Note: dt.entity.* is deprecated, but this is a migration-ASSESSMENT query — it inspects
// the classic constructs this migration replaces, so keep the classic query above:
//   - management zones have NO Smartscape node equivalent; they are what this migration
//     replaces with segments and policies.
//   - dt.security_context is an ARRAY on Smartscape nodes (empty [] when unset, not null),
//     so isNull / isNotNull(dt.security_context) does not carry over — a Smartscape rewrite
//     would miscount coverage.
//   - entity tags are not a flat "tags" field on Smartscape (resolve via getNodeField).
```

```
// Validate: Host security context coverage
fetch dt.entity.host
| summarize
    total = count(),
    withContext = countIf(isNotNull(dt.security_context)),
    missingContext = countIf(isNull(dt.security_context))
| fields total, withContext, missingContext,
         coveragePercent = round(100.0 * withContext / total, decimals: 2)

// Note: dt.entity.* is deprecated, but this is a migration-ASSESSMENT query — it inspects
// the classic constructs this migration replaces, so keep the classic query above:
//   - dt.security_context is an ARRAY on Smartscape nodes (empty [] when unset, not null),
//     so isNull / isNotNull(dt.security_context) does not carry over — a Smartscape rewrite
//     would miscount coverage.
```

Verify Security Context Distribution

```
// Distribution of security context values
// Verify expected teams/environments are represented
fetch dt.entity.service
| filter isNotNull(dt.security_context)
| summarize count = count(), by:{dt.security_context}
| sort count desc

// Note: dt.entity.* is deprecated, but this is a migration-ASSESSMENT query — it inspects
// the classic constructs this migration replaces, so keep the classic query above:
//   - dt.security_context is an ARRAY on Smartscape nodes (empty [] when unset, not null),
//     so isNull / isNotNull(dt.security_context) does not carry over — a Smartscape rewrite
//     would miscount coverage.
```

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3. Access Control Validation

Test User Access Scenarios

For each test scenario:
1. Log in as test user
2. Navigate to target app/data
3. Verify expected access
4. Document results

Scenario 1: Team-Based Access

Test User: Member of `frontend-team` group
Expected: Access to frontend services only

Validation Steps:
1. Open Services app
2. Select "Frontend Team" segment
3. Verify only frontend services visible
4. Try accessing backend service directly → Should fail

Scenario 2: Environment Restriction

Test User: Member of `prod-viewers` group
Expected: View production data only, no edit

Validation Steps:
1. Open Logs app
2. Select "Production" segment
3. Query logs → Should succeed
4. Try editing settings → Should fail
5. Select different segment → Should show no data (if boundary restricts)

Scenario 3: Full Access (SRE)

Test User: Member of `sre-team` group
Expected: Full access to production environment

Validation Steps:
1. Open any app
2. Verify data visible across all services
3. Verify edit capabilities work
4. Verify settings access

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4. Segment Validation

Verify Segment Data Match

Compare segment results with expected MZ results:

```
// Compare: MZ entity count vs Segment entity count
// Replace with your MZ name and segment filter
fetch dt.entity.service
| summarize
    viaMZ = countIf(in(managementZones, {"Frontend-Team"})),
    viaSegment = countIf(matchesValue(tags, "team:frontend"))
| fields viaMZ, viaSegment,
         match = if(viaMZ == viaSegment, then: "MATCH", else: "MISMATCH")

// Note: dt.entity.* is deprecated, but this is a migration-ASSESSMENT query — it inspects
// the classic constructs this migration replaces, so keep the classic query above:
//   - management zones have NO Smartscape node equivalent; they are what this migration
//     replaces with segments and policies.
//   - entity tags are not a flat "tags" field on Smartscape (resolve via getNodeField).
```

```
// Find entities in MZ but not matching segment filter
// These are migration gaps
fetch dt.entity.service
| filter in(managementZones, {"Frontend-Team"})
| filter NOT matchesValue(tags, "team:frontend")
| fields entity.name, tags, managementZones

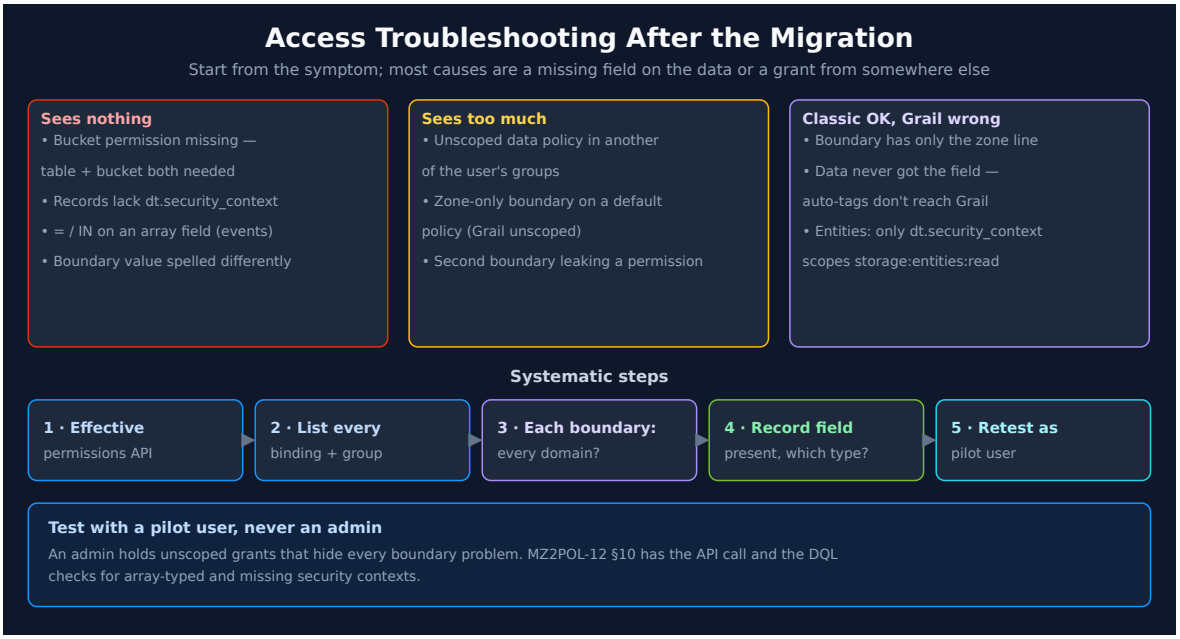
// Note: dt.entity.* is deprecated, but this is a migration-ASSESSMENT query — it inspects
// the classic constructs this migration replaces, so keep the classic query above:
//   - management zones have NO Smartscape node equivalent; they are what this migration
//     replaces with segments and policies.
//   - entity tags are not a flat "tags" field on Smartscape (resolve via getNodeField).
```

```
// Find entities matching segment but not in MZ
// May indicate over-inclusion in new model
fetch dt.entity.service
| filter matchesValue(tags, "team:frontend")
| filter NOT in(managementZones, {"Frontend-Team"})
| fields entity.name, tags, managementZones

// Note: dt.entity.* is deprecated, but this is a migration-ASSESSMENT query — it inspects
// the classic constructs this migration replaces, so keep the classic query above:
//   - management zones have NO Smartscape node equivalent; they are what this migration
//     replaces with segments and policies.
//   - entity tags are not a flat "tags" field on Smartscape (resolve via getNodeField).
```

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5. Common Issues and Troubleshooting



Issue 1: User Can't See Any Data

Symptoms:
- Apps show empty results
- DQL queries return no records

Troubleshooting:

Check	Action
Group membership	Verify user is in correct group
Policy binding	Verify group has policy bound
Boundary scope	Verify boundary isn't too restrictive
Data policy	Verify a data-read policy (`Read Logs` / `All Grail data read access`) is included
Segment selection	Check if restrictive segment is active

Resolution Steps:
1. Check user's group membership in Account Management
2. Verify group has appropriate policy binding
3. Review boundary conditions
4. Test with admin account to verify data exists

Issue 2: User Sees Too Much Data

Symptoms:
- User sees data outside their scope
- Cross-team data visible

Troubleshooting:

Check	Action
Boundary conditions	Verify boundary is correctly restrictive
Multiple group membership	User may be in multiple groups
Security context	Verify entities have correct context
Policy conditions	Check for overly broad conditions

Resolution Steps:
1. Review all groups user belongs to
2. Check all policy bindings for those groups

3. Verify boundary conditions are correct
4. Check security context on visible entities

Issue 3: Segment Filter Not Working

Symptoms:
- Segment selected but data not filtered
- Same data visible with/without segment

Troubleshooting:

Check	Action
Segment syntax	Verify DQL filter is correct
Tag existence	Verify entities have expected tags
App support	Verify app supports segments
Segment sharing	Verify segment is shared with user

Resolution Steps:
1. Test segment filter as standalone DQL query
2. Verify tags exist on target entities
3. Check segment is shared/public
4. Try segment in different app

Issue 4: Classic Apps Work, New Apps Don't

Symptoms:
- Legacy UI shows expected data
- New apps (Services, Logs) show no/wrong data

Cause: MZ-based access works for classic, but new apps need segments

Resolution:
1. Create corresponding segment for MZ filter
2. Ensure user selects segment in new apps
3. Consider setting default segment for user/group

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6. Diagnostic Queries

Check Entity Tag Distribution

```
// Analyze tag patterns for segment creation
fetch dt.entity.service
| expand tag = tags
| parse tag, "LD:key ':' LD:value"
| summarize count = count(), by:{key}
| sort count desc
| limit 20

// Note: dt.entity.* is deprecated, but this is a migration-ASSESSMENT query — it inspects
// the classic constructs this migration replaces, so keep the classic query above:
//   - entity tags are not a flat "tags" field on Smartscape (resolve via getNodeField).
```

```
// Find entities without standard tags
// These may need tagging for segment filters
fetch dt.entity.service
| filter NOT matchesValue(tags, "team:*")
| filter NOT matchesValue(tags, "env:*")
| fields entity.name, tags
| limit 50

// Note: dt.entity.* is deprecated, but this is a migration-ASSESSMENT query — it inspects
// the classic constructs this migration replaces, so keep the classic query above:
//   - entity tags are not a flat "tags" field on Smartscape (resolve via getNodeField).
```

Check MZ Assignment Health

```
// Entities with MZ but no security context
// These need security context for boundary-based access
fetch dt.entity.service
| filter arraySize(managementZones) &gt; 0
| filter isNull(dt.security_context)
| fields entity.name, managementZones, tags
| limit 50

// Note: dt.entity.* is deprecated, but this is a migration-ASSESSMENT query — it inspects
// the classic constructs this migration replaces, so keep the classic query above:
//   - management zones have NO Smartscape node equivalent; they are what this migration
//     replaces with segments and policies.
//   - dt.security_context is an ARRAY on Smartscape nodes (empty [] when unset, not null),
//     so isNull / isNotNull(dt.security_context) does not carry over — a Smartscape rewrite
//     would miscount coverage.
//   - entity tags are not a flat "tags" field on Smartscape (resolve via getNodeField).
```

```
// MZ to tag alignment check
// Verify MZ membership aligns with tags used in segments
fetch dt.entity.service
| expand mz = managementZones
| summarize {
    entityCount = count(),
    withTeamTag = countIf(matchesValue(tags, "team:*")),
    withEnvTag = countIf(matchesValue(tags, "env:*"))
  }, by:{mz}
| sort entityCount desc

// Note: dt.entity.* is deprecated, but this is a migration-ASSESSMENT query — it inspects
// the classic constructs this migration replaces, so keep the classic query above:
//   - management zones have NO Smartscape node equivalent; they are what this migration
//     replaces with segments and policies.
//   - entity tags are not a flat "tags" field on Smartscape (resolve via getNodeField).
```

7. Ongoing Maintenance

Regular Validation Tasks

Task	Frequency	Purpose
Security context audit	Weekly	Ensure new entities have context
Segment effectiveness	Monthly	Verify segments return expected data
Access review	Quarterly	Verify user access is appropriate
Policy review	Quarterly	Ensure policies match requirements

New Entity Onboarding

When new services/hosts are deployed:

1. **Auto-tagging**: Ensure auto-tagging rules apply correct tags
2. **Security context**: Verify security context is set
3. **Segment inclusion**: Verify entity appears in appropriate segments
4. **Access verification**: Test user access to new entity

New User Onboarding

When new users join:

1. Add user to appropriate group(s)
2. Communicate available segments
3. Provide access documentation
4. Verify access works as expected

8. Health Monitoring Queries

Create Monitoring Dashboard

Use these queries for an access health dashboard:

```
// KPI: Security Context Coverage
fetch dt.entity.service
| summarize
    total = count(),
    covered = countIf(isNotNull(dt.security_context))
| fields coveragePercent = round(100.0 * covered / total, decimals: 1)

// Note: dt.entity.* is deprecated, but this is a migration-ASSESSMENT query — it inspects
// the classic constructs this migration replaces, so keep the classic query above:
//   - dt.security_context is an ARRAY on Smartscape nodes (empty [] when unset, not null),
//     so isNull / isNotNull(dt.security_context) does not carry over — a Smartscape rewrite
//     would miscount coverage.
```

```
// KPI: Entities per Security Context
// Monitor for unexpected growth/decline
fetch dt.entity.service
| summarize count = count(), by:{dt.security_context}
| sort count desc
| limit 10

// Note: dt.entity.* is deprecated, but this is a migration-ASSESSMENT query — it inspects
// the classic constructs this migration replaces, so keep the classic query above:
//   - dt.security_context is an ARRAY on Smartscape nodes (empty [] when unset, not null),
//     so isNull / isNotNull(dt.security_context) does not carry over — a Smartscape rewrite
//     would miscount coverage.
```

```
// KPI: Untagged Entities (migration gaps)
fetch dt.entity.service
| filter isNull(dt.security_context)
| filter NOT matchesValue(tags, "team:*")
| summarize untaggedCount = count()

// Note: dt.entity.* is deprecated, but this is a migration-ASSESSMENT query — it inspects
// the classic constructs this migration replaces, so keep the classic query above:
//   - dt.security_context is an ARRAY on Smartscape nodes (empty [] when unset, not null),
//     so isNull / isNotNull(dt.security_context) does not carry over — a Smartscape rewrite
//     would miscount coverage.
//   - entity tags are not a flat "tags" field on Smartscape (resolve via getNodeField).
```

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9. Best Practices Summary

Access Control

- ✅ Use groups, not individual user permissions
- ✅ Apply least privilege principle
- ✅ Document all policy bindings
- ✅ Regular access reviews

Segments

- ✅ Align with organizational structure
- ✅ Use consistent naming conventions
- ✅ Test filters before deployment
- ✅ Share segments appropriately

Boundaries

- ✅ Create reusable boundaries

- ✅ Use security context for entity-level control
- ✅ Document boundary purpose
- ✅ Keep conditions simple

Maintenance

- ✅ Automate security context assignment
- ✅ Monitor access health metrics
- ✅ Review and update documentation
- ✅ Train new team members

Summary

In this notebook, you learned:

1. **Validation procedures**: How to verify migration success
2. **Troubleshooting**: Common issues and resolutions
3. **Diagnostic queries**: DQL for investigating access problems
4. **Ongoing maintenance**: Regular tasks for access health
5. **Health monitoring**: Queries for access dashboards

Migration Complete!

Congratulations on completing the MZ2POL migration series. You have:

- ✅ Understood why migration is necessary (MZ2POL-01)
- ✅ Learned the new access control model (MZ2POL-02)
- ✅ Assessed and planned your migration (MZ2POL-03)
- ✅ Configured policies and boundaries (MZ2POL-04)
- ✅ Implemented segments (MZ2POL-05)
- ✅ Executed the migration (MZ2POL-06)
- ✅ Validated and can troubleshoot (MZ2POL-07)
- ✅ Templated policies for scale (MZ2POL-08)

Additional Resources

- Access Management Concepts
- Dynatrace Community - Boundaries + Policies Best Practices
- Segments Documentation
- IAM Policy Reference
- Policy Templating

This notebook was AI-generated from community-submitted and publicly available sources. This notebook series is not officially supported by Dynatrace. Always verify information against official Dynatrace documentation.